Steven Buchanan

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Australian National University PhD Candidate · expected completion Feb 2027

Education

Australian National University: PhD in Mathematics — Sept 2023 – est. Feb 2027
– Research in geometric analysis and nonlinear PDE
– Visiting PhD Researcher, University of Toronto, Aug 2025 – Jan 2026

University of Tennessee: PhD studies in Mathematics — Aug 2021 – Sept 2023
– transferred to ANU

Tennessee Technological University: M.S. in Mathematics — Aug 2019 – May 2021

Tennessee Technological University: B.S. in Mathematics — Aug 2015 – May 2019

Projects

Global material stock database and budget – Machine learning research team

- Contribute to development of a harmonized global database of building-material stocks for climate and resource-budget modelling.
- Apply machine-learning and statistical methods to heterogeneous research datasets, including model development, validation, and quantitative analysis.
- Work collaboratively on computational research involving data cleaning, modelling, and reproducible analysis in Python.

Market-Implied Stock Price Densities – Erdős Institute certification project

- Derived and implemented Breeden–Litzenberger density extraction to estimate market-implied SPX risk-neutral distributions from cross-sectional option prices.
- Designed a no-arbitrage constrained option-price fitting procedure, enforcing decreasing and convex call-price curves before numerically estimating second derivatives.
- Backtested ~6,900 option-implied distributions against realized SPX expirations using CRPS, finding the constrained price-fitting approach reduced average relative CRPS from ~0.0401 to ~0.0345 versus an implied-volatility smoothing approach.
- Investigated the equity pricing kernel by transforming risk-neutral densities under CRRA utility and numerically estimating the risk-aversion parameter that minimized historical distributional forecast error.

Deformation of Surfaces by Functions of Curvature – Doctoral Research

- Research nonlinear geometric evolution equations generalizing Ricci/Yamabe flow, using techniques from differential geometry, nonlinear parabolic PDEs, and asymptotic analysis.
- Analyze existence, singularity formation, and long-time behavior of curvature-driven systems; establish estimates leading to convergence of compact non-negatively curved surfaces toward round geometries under rescaling.

- Develop and apply evolution equations, maximum principles, differential Harnack inequalities, entropy/monotonicity methods, and scaling arguments to control nonlinear PDE solutions.

Asymptotic Analysis of Arithmetic Functions – Master's Thesis

- Derived asymptotic formulae for sums of classes of divisor functions over arithmetic progressions using methods from analytic number theory.

Probabilistic modeling of divisibility patterns in integers – Undergraduate research project

- Conducted asymptotic and probabilistic analysis of divisibility relations to quantify how often a generalized version of Euclid's lemma holds under random integer sampling.
- Research published in *The American Mathematical Monthly*.

Skills

Programming & data analysis: Python, NumPy, Pandas, SciPy, Matplotlib, SQL, Jupyter, Excel
Machine learning & quantitative methods: PyTorch, regression analysis, random forests, gradient boosting, hyperparameter tuning, probability and statistics, stochastic processes, time-series analysis, optimisation, Monte Carlo simulation
Tools & technologies: Git, command–line Linux, LaTeX, TypeScript

Teaching

Instructor and teaching assistant for differential geometry, differential equations, linear algebra, discrete mathematics, and college algebra at ANU, Shandong University, University of Tennessee and Tennessee Tech. Designed lectures, assessments and problem-solving tutorials for undergraduate students.

Awards

HDR Fee Remission Merit Scholarship and International PhD scholarship – Australian National University
F. M. Dryzer Math Fellowship – University of Tennessee